

EDA Summary Report: Geldium Dataset

1. Project Introduction

This report covers the Exploratory Data Analysis (EDA) I performed on the Geldium dataset. The primary goal was to check the data quality before moving on to building a delinquency prediction model. I specifically looked for missing data, potential errors, and how different variables might relate to a customer's risk of defaulting.

2. A Look at the Dataset

After an initial review of the 500 customer records, the data seemed fairly solid. I checked the structure for formatting issues and looked for extreme outliers that did not make sense. Aside from a few missing values, it was a clean dataset.

Key details include:

- Total Records: 500 customers
- Variables: Income, Credit Score, Loan Balance, Credit Utilization, Missed Payments, Account Tenure, and a six-month payment history (Month_1 to Month_6), along with the target variable Delinquent_Account.
- Data Types: A mix of numerical, categorical, and binary variables.

3. Dealing with Missing Values

I found 70 missing values in total, mostly concentrated in a few columns.

- Income: 39 missing (7.8%)
- Loan_Balance: 29 missing (5.8%)
- Credit_Score: 2 missing (0.4%)

To handle these missing values, I used median imputation. I chose the median instead of the mean because it is less affected by extreme values and provides a more reliable estimate for financial data.

4. What the Data Tells Us (Key Findings)

I performed a correlation analysis to identify variables associated with delinquency.

The results showed that no single variable had a strong correlation with Delinquent_Account. The strongest relationships, although still weak, were observed for Income, Credit Utilization, and the first month of payment history.

Overall, the findings suggest that delinquency is influenced by a combination of financial and behavioral factors rather than one single cause. No significant anomalies or data quality issues were identified during the analysis.

5. Use of Generative AI

Generative AI was used to help explore possible approaches for handling missing values and to assist in summarizing the analytical findings. The final preprocessing decisions and interpretations were based on the characteristics of the dataset.

Example prompts:

- "What are the best ways to handle missing income data in a small financial dataset?"
- "Help me summarize these correlation results for a college report."
- "Explain the relationship between credit utilization and default risk."

6. Summary and Moving Forward

The dataset has now been prepared for predictive modeling, and the missing values have been addressed. The main takeaway is that delinquency is a complex outcome influenced by multiple factors rather than a single variable.

Next, feature engineering will be performed, followed by the development and evaluation of models such as Logistic Regression and Random Forest to improve delinquency prediction.